JFOX-FMU v1
STM32H753IIT6 flight controller – three IMUs, two barometers,
CAN FD with jumpered termination, 1 MB RAM.

Generated by hardware/tools/gen_fmu_schematic.py – do not hand-edit, regenerate. Pin assignment comes from the part's own alternate-function table (PINMAP.md); sensor pinouts from the datasheets (PINOUTS.md).

Cross-sheet nets are global labels, which is safe here because this is one board – unlike the 3-board TMR project, where a global would short all three instances together.

MCU

File: mcu.kicad_sch

SENSORS

File: sensors.kicad_sch

POWER

File: power.kicad_sch

COMMS

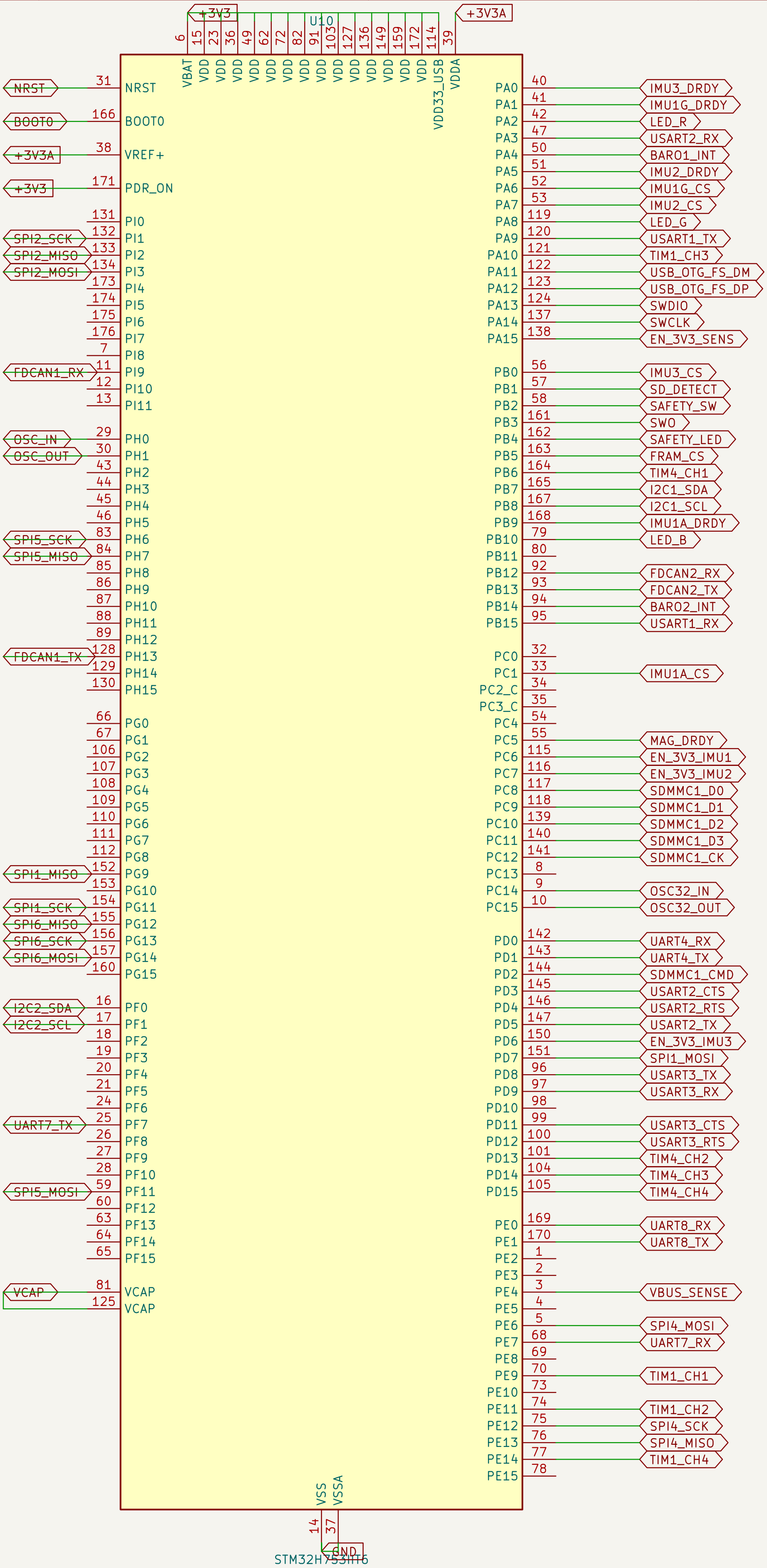
File: comms.kicad_sch

PASSIVES

File: passives.kicad_sch

Sheet 1 of 6
Generated by hardware/tools/gen_fmu_schematic.py – regenerate, do not edit
STM32H753IIT6 – 3 IMUs, 2 barometers, magnetometer, CAN FD
JFOX Aircraft Co., Ltd.
Sheet: /
File: jfox-fmu.kicad_sch
Title: JFOX-FMU v1

Size: A4	Date: 2026-08-03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/6



VCAP1/VCAP2 are the internal LDO output. NOT a supply — do not tie to +3V3

Pins allocated from the part's own AF table — see PINMAP.md

STM32H753IIT6 LQFP176 — 1 MB RAM, 2 MB flash, Cortex-M7 480 MHz

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Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU

Size: A4	Date: 2026-08-03	**Rev: A**
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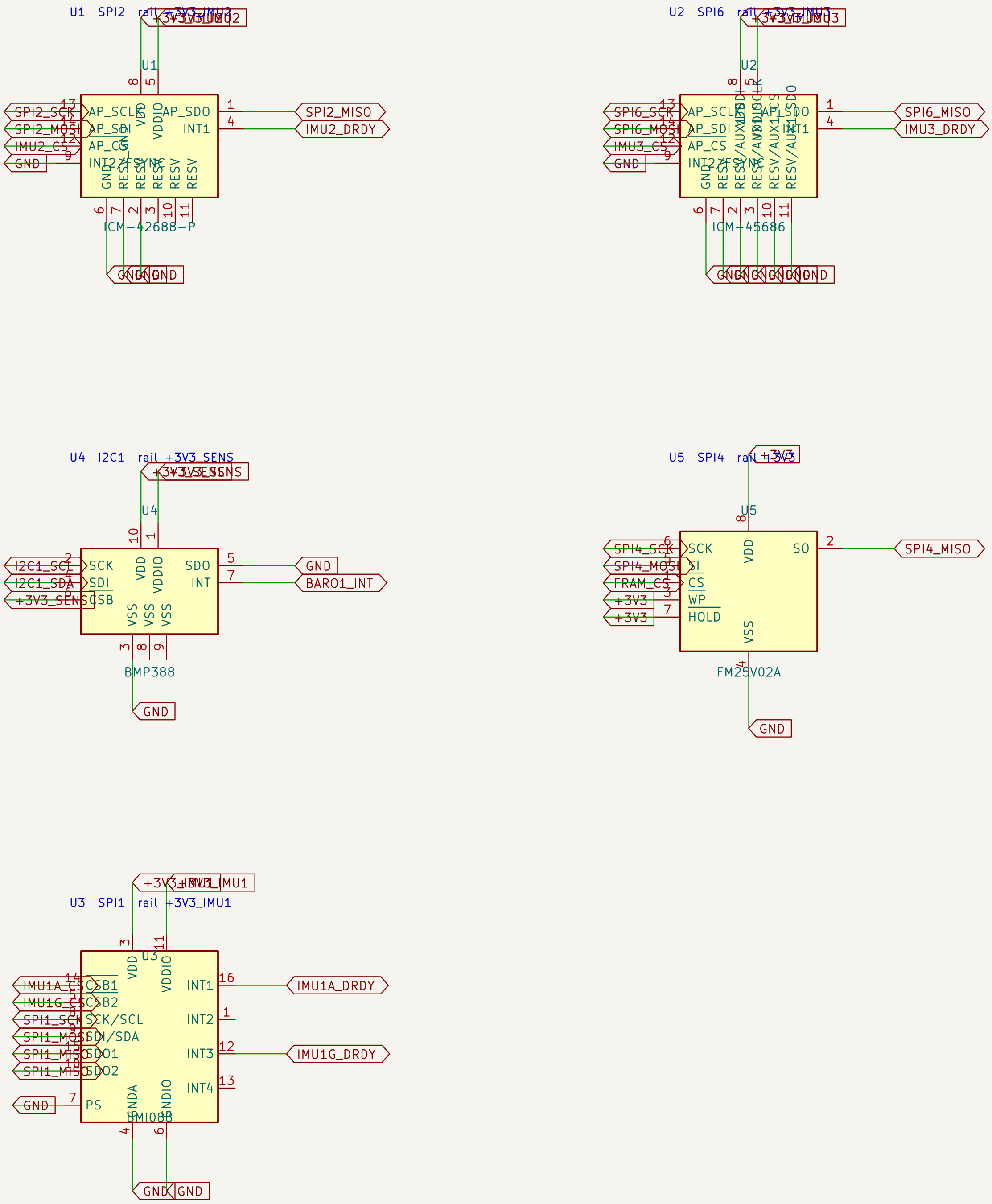
Three IMUs on three separate SPI buses, each on its own switchable rail. One sensor hanging its bus must not take the other two with it.

WARNING — before dropping any +3V3_IMUn rail, firmware must drive that bus's SCK/MOSI/CS low. Interface pins held high with VDDIO off destroy these parts through their ESD diodes (BMP388 datasheet 3.2).



Drive a bus low BEFORE dropping its rail — see ARCHITECTURE.md

Three IMUs, three separate SPI buses, three switchable rails

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Sheet: /SENSORS/

File: sensors.kicad_sch

Title: Sensors

Size: A4	Date: 2026-08-03	**Rev: A**
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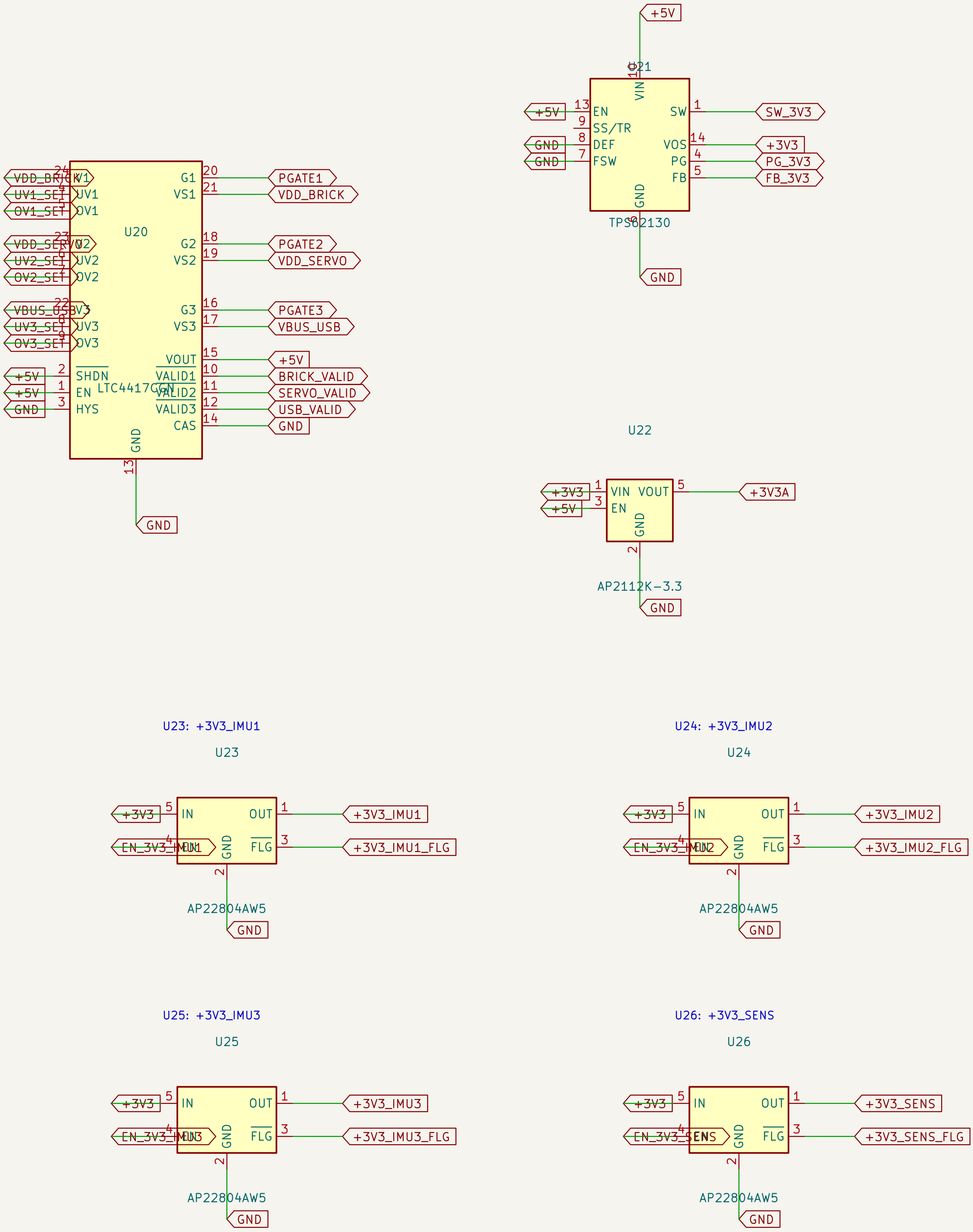
LTC4417 prioritised ORing: brick > servo rail > USB, with under- and over-voltage lockout per input. This is v2.4.5's arrangement kept, correctly identified – that board's docs called it a BQ24315 until the netlist proved otherwise.

U21 is a BUCK, not an LDO. At 5V in, 3V3 out and the measured ~310 mA typical load an LDO burns 0.53 W, and 0.94 W at peak, which an SOT-23-5 cannot shed – see POWER_BUDGET.md. U22 stays an LDO because it carries only VDDA/VREF+, a few mA, and its quietness is the point.

One load switch per sensor bus, so a wedged IMU can be power-cycled without disturbing the other two.

WARNING – firmware must drive a bus's SCK/MOSI/CS low BEFORE clearing its EN_3V3_* line. Interface pins held high with the rail down destroy these sensors through their ESD diodes (BMP388 datasheet 3.2).



One switchable rail per sensor bus
U21 is a buck (POWER_BUDGET.md); U22 is an LDO for the analog rail
LTC4417 prioritised ORing: brick > servo > USB

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Sheet: /POWER/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-08-03	**Rev: A**
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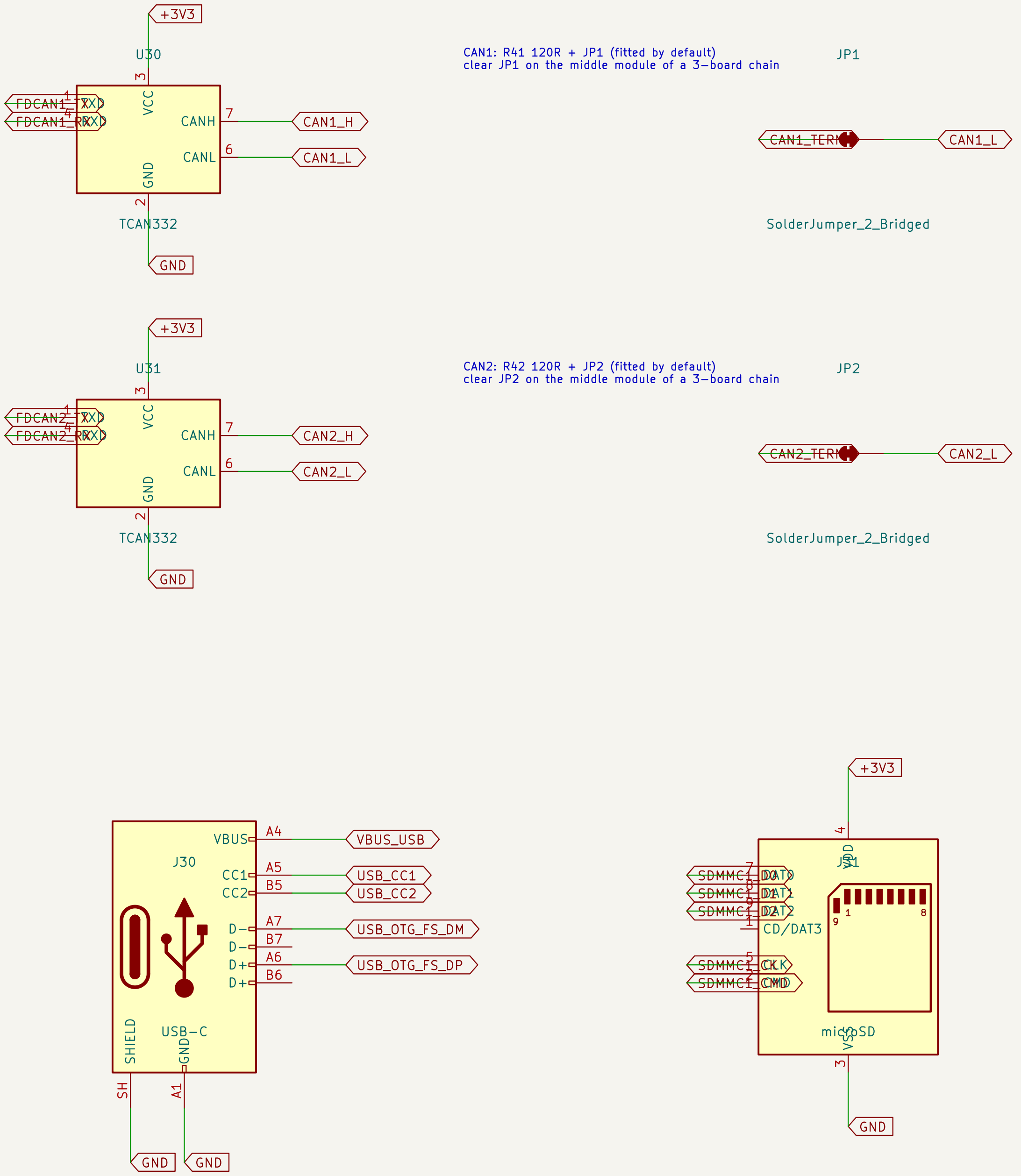
TWO CAN FD BUSES, BOTH WITH TRANSCEIVERS. v2.4.5 wired CAN2 to the MCU
with no transceiver at all, so it was never usable as a bus.

TERMINATION IS SWITCHABLE. 120 ohm in series with a solder jumper,
FITTED by default. On a three-board TMR chain the electrically middle
module clears JP1 — it does not get R409 desoldered, which is what the
old board required. Verify ~60 ohm across CAN_H/CAN_L, bus unpowered,
before trusting it.



Termination is jumpered: clear JPn on the middle module of a chain
Two CAN FD buses, both with transceivers — v2.4.5 had only one
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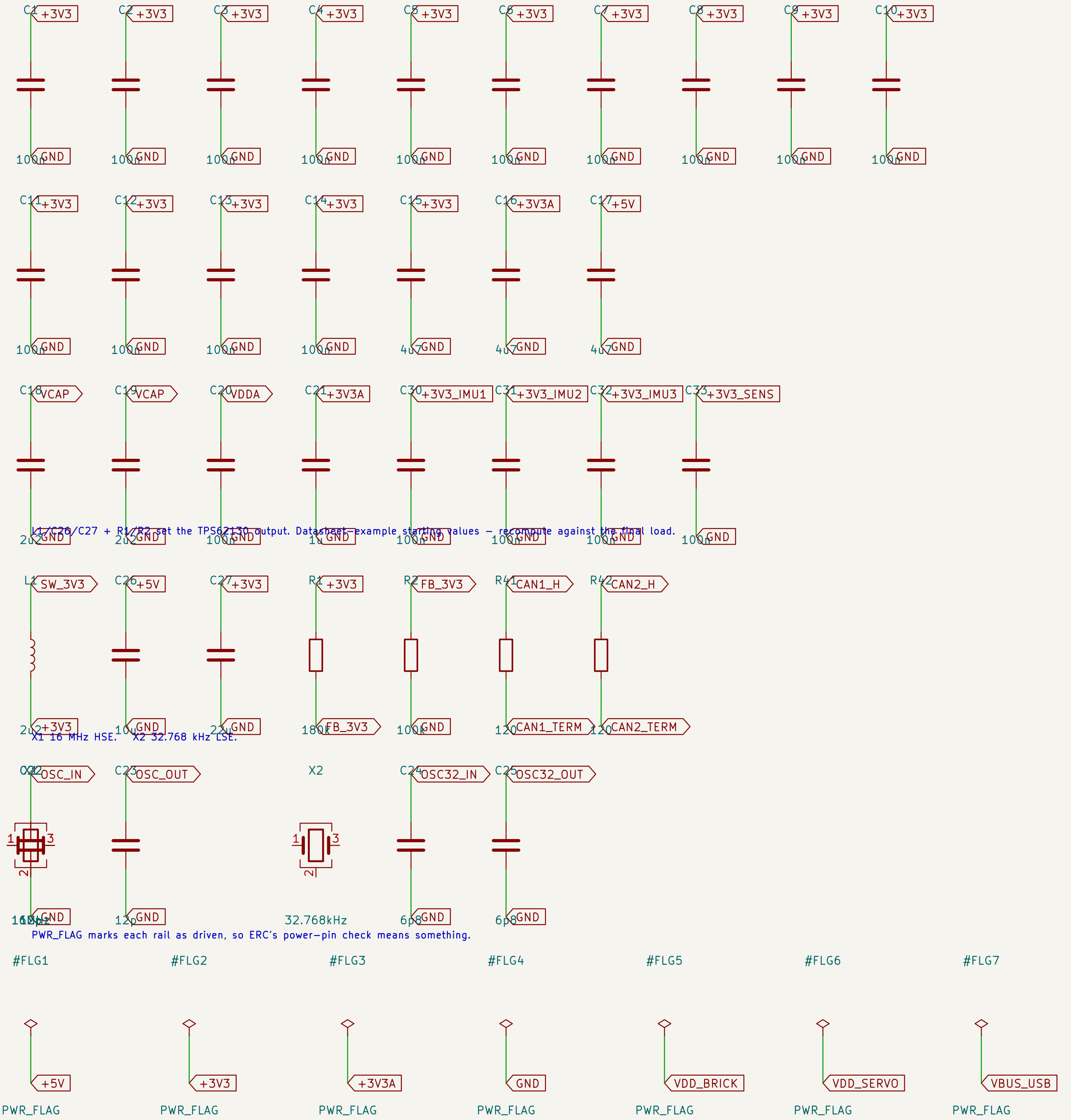
Sheet: /COMMS/
File: comms.kicad_sch
Title: Comms and Storage
Size: A4 | Date: 2026-08-03 | **Rev: A**
KiCad E.D.A. 10.0.5 | Id: 5/6

Decoupling: one 100n per MCU VDD pin (14), plus 4u7 bulk per rail.
VCAP1/VCAP2 get 2u2 each — the internal LDO's output, not a supply.

16 MHz HSE divides exactly to 48 MHz for USB. The old board's 24 MHz could not, which left its USB clock at 51.4 MHz.

CAN terminators are HERE, in series with the comms-sheet jumpers.



CAN terminators R41/R42 live here, not on the comms sheet
One 100n per MCU VDD pin, 2u2 per VCAP, bulk per rail

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Sheet: /PASSIVES/
File: passives.kicad_sch

Title: Passives and Clocks

Size: A4	Date: 2026-08-03	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 6/6